

AI Engineer Intern POC doc

 Reference Material:

https://dour-blade-da4.notion.site/Compra-Assignment-Reference-Document-361573730b5a804aa76df3c680e6cae0?source=copy_link

Please build a chat-based layout agent.

We have provided a design JSON (at the end of this doc) that contains the canvas, layers, text, image URLs, positions, sizes, and semantic roles.

The user should be able to chat with the agent and ask things like:

Convert this design to 9:16
Keep the product large
Move the headline to the top
Move the offer badge higher
Make the headline smaller

The agent should update the layout JSON based on the instruction and show the updated JSON in the UI.

No PSD parsing is required.
No final PNG rendering is required.
A basic wireframe preview is optional.

We are evaluating:

Chat interface
LLM integration
Layout reasoning
JSON transformation
Ability to handle follow-up instructions

Please submit:

GitHub repo
README with setup instructions
Short note explaining your approach
Optional Loom walkthrough

Time limit: 24 hours.

JSON below

```
{
  "rootNodes": [
    "artboard_1778485662755_3"
  ],
  "imageUrl":
  "https://res.cloudinary.com/dzydbgyfj/image/upload/v1778502510/Instagram_Post_1080x1080_19_iw9uqy.png",
  "nodes": {
    "img_1778485681535_4": {
      "x": -2.480199702295863,
      "y": 0,
      "id": "img_1778485681535_4",
      "nh": 1,
      "nw": 1,
      "nx": -0.0022964812058295027,
      "ny": 0,
      "data": {
        "fit": "cover",
        "assetId": "asset_1778485681534_jl4fd6",
        "sourceUrl": "https://res.cloudinary.com/dzydbgyfj/image/upload/v1778485682/qgi9ecexyirh7vbvij9k.png"
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      "name": "Background.png",
      "type": "image",
      "style": {
        "layout": {},
        "visual": {
          "borderRadius": 0
        }
      },
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    "parentId": "artboard_1778485662755_3"
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  "img_1778486846247_10": {
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    "y": 124.94234803158162,
    "id": "img_1778486846247_10",
    "nh": 0.037741738009449737,
    "nw": 0.034185517608088176,
    "nx": 0.30870529955458104,
    "ny": 0.11568735928850149,
    "data": {
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      "assetId": "asset_1778486846244_9hw86o",
      "sourceUrl": "https://res.cloudinary.com/dzydbgyfj/image/upload/v1778486847/c2pgksvxa3ec61bchter.png"
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    "name": "Vector (2).png",
    "type": "image",
    "style": {
      "layout": {},
      "visual": {
        "borderRadius": 0
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  },
  "width": 36.920359016735226,
```

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"height": 40.761077050205714,
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  "y": 126.70185327118239,
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  "nh": 0.03774173800944972,
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  "ny": 0.11731653080665036,
  "data": {
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    "assetId": "asset_1778486856821_s8v1en",
    "sourceUrl": "https://res.cloudinary.com/dzydbgyfj/image/upload/v1778486857/ksyoela4vnnvrtlt49lf1.png"
  },
  "name": "Vector (1).png",
  "type": "image",
  "style": {
    "layout": {},
    "visual": {
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  "width": 36.920359016735226,
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  "parentId": "artboard_1778485662755_3"
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  "y": 127.86270704831685,
  "id": "img_1778487081392_12",
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  "nx": 0.2627754363955789,
  "ny": 0.1183913954151082,
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    "assetId": "asset_1778487081391_spprg9",
    "sourceUrl": "https://res.cloudinary.com/dzydbgyfj/image/upload/v1778487081/y6sbozdszgf0hc9wjmev.png"
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  "nx": 0.3203899861054117,
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  "nx": 0.17998798648142267,
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"ny": 0.32422264476723023,
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  "content": "Comfort that defines modern living."
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    "visual": {
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  "data": {
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  "nw": 0.16302282390535613,
  "nx": 0.12118077142907656,
  "ny": 0.415289376333061,
  "data": {
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  },
  "name": "Text",
  "type": "text",
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    "layout": {},
    "visual": {
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  "id": "circle_1778488914968_15",
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  "nw": 0.14789099725854568,
  "nx": 0.09601711880487064,
  "ny": 0.4050170159100531,
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  "type": "shape",
  "style": {
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        "value": "#F4CF1B"
      },
      "stroke": {
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        "value": "#F4CF1B"
      },
      "strokeWidth": 2
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  },
  "width": 159.72227703922934,
  "height": 153.88155900575885,
  "parentId": "artboard_1778485662755_3"
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"artboard_1778485662755_3": {
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  "y": -724.4991007769831,
  "id": "artboard_1778485662755_3",
  "data": {
    "preset": "instagram-post",
    "showLabel": true,
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  },
  "name": "Instagram Post",
  "type": "artboard",
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"style": {
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"children": [
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  "img_1778487081392_12",
  "img_1778487101466_13",
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  "text_1778489078397_16",
  "img_1778489515746_17"
]
}
}
}
```